

Foundation Program in Literacy, Numeracy and Skills

NUMERACY

Teacher's Guide: GRADE 9 –Lesson 8

Volume in real life

Volume is used as a measurement of the space occupied by an object and the units of measurement are cm^3 and m^3 .



A. Volume of Cubes and Cuboids

B. Competency, Element and Performance Criteria:

C 2.3: Use measurement in real-life situations

Element 6: Applying concept of volume in real-life

Performance Criteria:

Level 1: Find volume of cubes and cuboids.

Level 2: Applying concept of volume in real life.

C. Purpose of activity:

Students will understand and apply the formula for calculating volume of cubes and cuboids in practical scenarios related to different professions.

D. Learning Outcomes:

By the end of the lesson, students should be able to

- Understand the concept of volume.
- Apply formulas for calculating the volume of cubes and cuboids.
- Relate volume calculations to real-world contexts (e.g., packaging, construction, storage)

E. Resources and materials:

- Graph paper, rulers, cubes, and cuboid models.
- Worksheets for calculations.
- Digital tools (GeoGebra or 3D modeling apps, if available).

F. Teaching trajectories/Implementation guidelines:

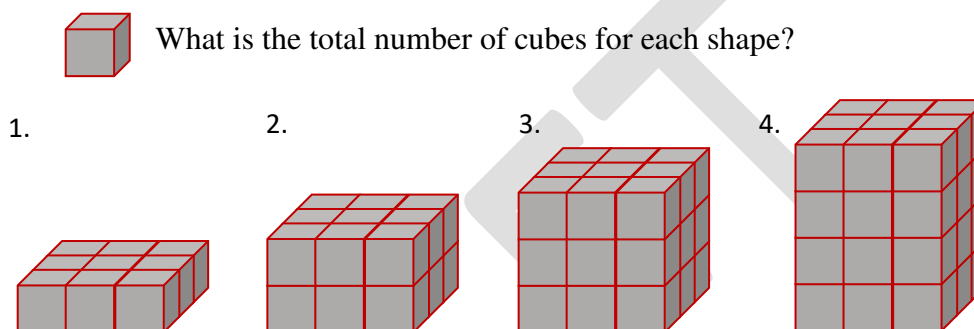
Activity 1: “Build-a-Block: Cubes and Cuboids”

(Teacher asks students to bring cubical playing dice and encourages them to construct cubes and cuboids by using the dice.)

Objective:

Students will:

- Understand the structure of cubes and cuboids.



Activity 2: Measuring the space occupied by a cube using unit cubes

Objective:

Students will:

- Identify dimensions (length, width, height) of constructed shapes.
- Explore volume of a cube.

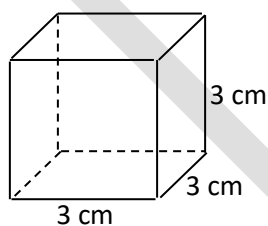


Figure 1

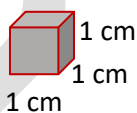


Figure 2

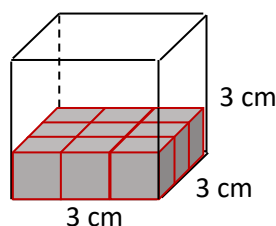


Figure 3

If we fill the 3-cm cube (Figure 1) completely with the 1-cm cubes (Figure 2).

- (a) How many cubes are there along the length of Figure 3?
- (b) How many cubes are there along the width of Figure 3?
- (c) How many cubes are there along the height of Figure 3?
- (d) How many 1-cm cubes fill the space in the cube in Figure 3 completely?

The **volume** of a cube is the amount of space that it occupies. In this example 27 unit cubes (1 cm^3) occupy the whole space of the cube in Figure 3. We say that the cube has a **volume** of 27 cm^3 (Figure 4).

Volume of cube = length $\times$ width $\times$ height

$$= 3\text{cm} \times 3\text{cm} \times 3\text{cm}$$

$$= 27\text{ cm}^3$$

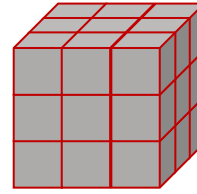
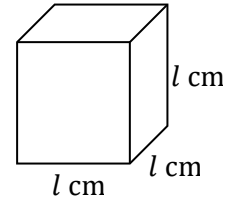


Figure 4

The volume a cube of length l cm is calculated as follows:

$$\text{Volume of cube} = l\text{cm} \times l\text{cm} \times l\text{cm}$$

$$= l^3\text{cm}^3$$



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Volume of Cubes and Cuboids

Activity 4:

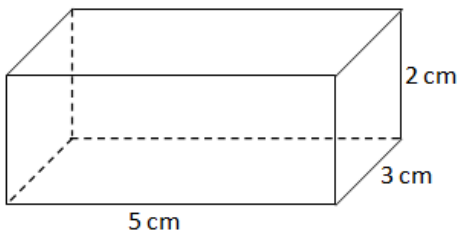


Figure 1

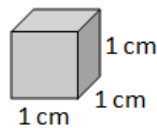


Figure 2

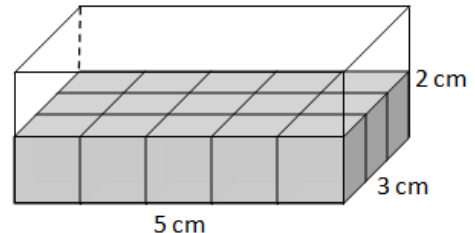
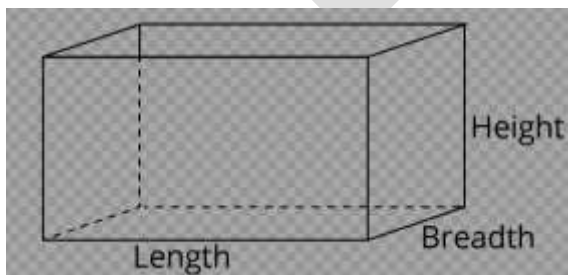


Figure 3

- (a) What is the name the shape in Figure 1?
If we fill the shape in Figure 1 completely with the 1-cm cubes (Figure 2).
- (b) How many 1-cm cubes fill the space in the cube in Figure 3 completely?
- (c) What is the volume of the shape in Figure 1?

The Cuboid



The volume of a cuboid is given by:

$$\text{Volume of cuboid} = l \times b \times h ,$$

where l represents the length, b represents the breadth and h represents the height of the cuboid.